Laboratory Work 5

Spring Boot Event Registration API

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Purpose: Build an in-memory REST backend using Spring Boot, a service layer, constructor-based dependency injection, and settings from application.properties.

Implementation

The project uses Gradle and Java 17-compatible source code. RegistrationController exposes JSON GET endpoints and delegates event data to EventService through constructor injection. EventInfo is the JSON model; the service keeps three events in memory. The /api/info response reads the application name, course, and semester from application.properties. The optional event lookup returns a clear JSON 404 response when an ID is not found.

Validation completed: Gradle tests passed, the application started on port 8080, and all required endpoints plus the missing-event response were checked in a browser and with HTTP requests.

Endpoint screenshots

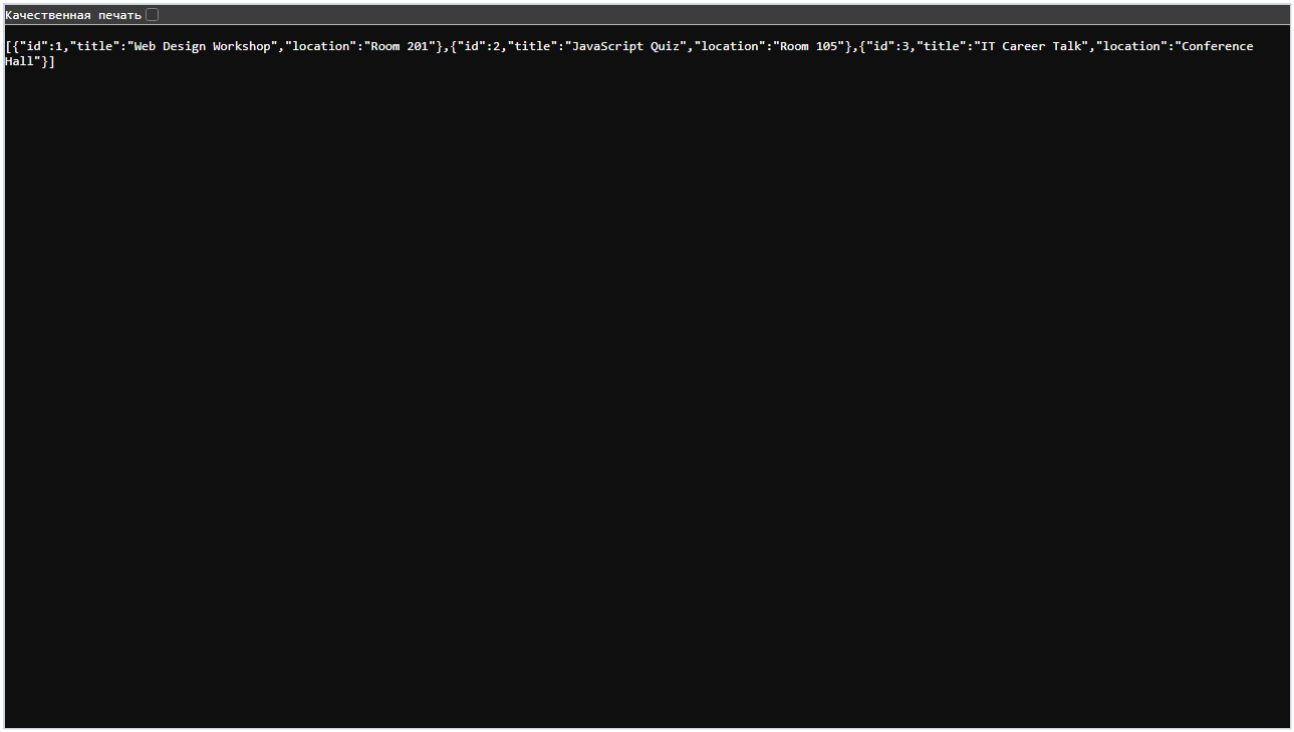


Figure 1. GET /api/events — three event objects returned as JSON.

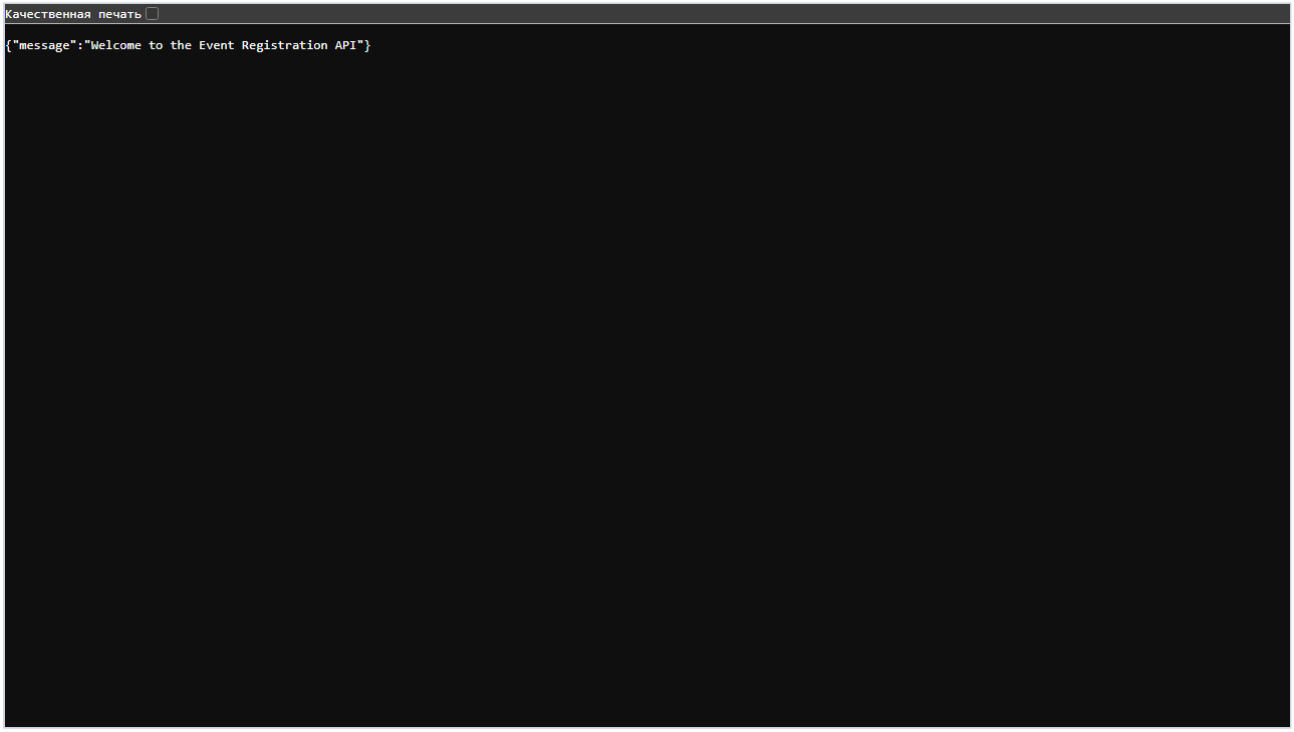


Figure 2. GET /api/hello — the service welcome message.

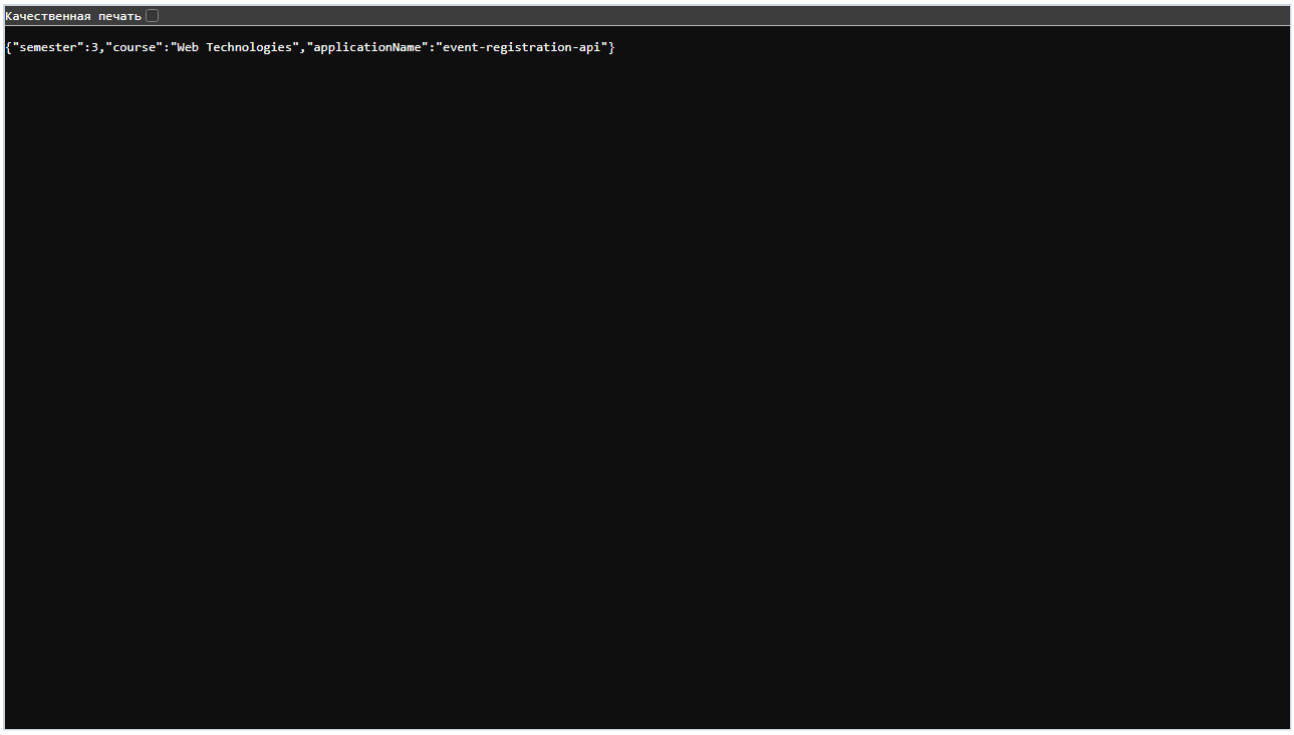


Figure 3. GET /api/info — application settings read from application.properties.

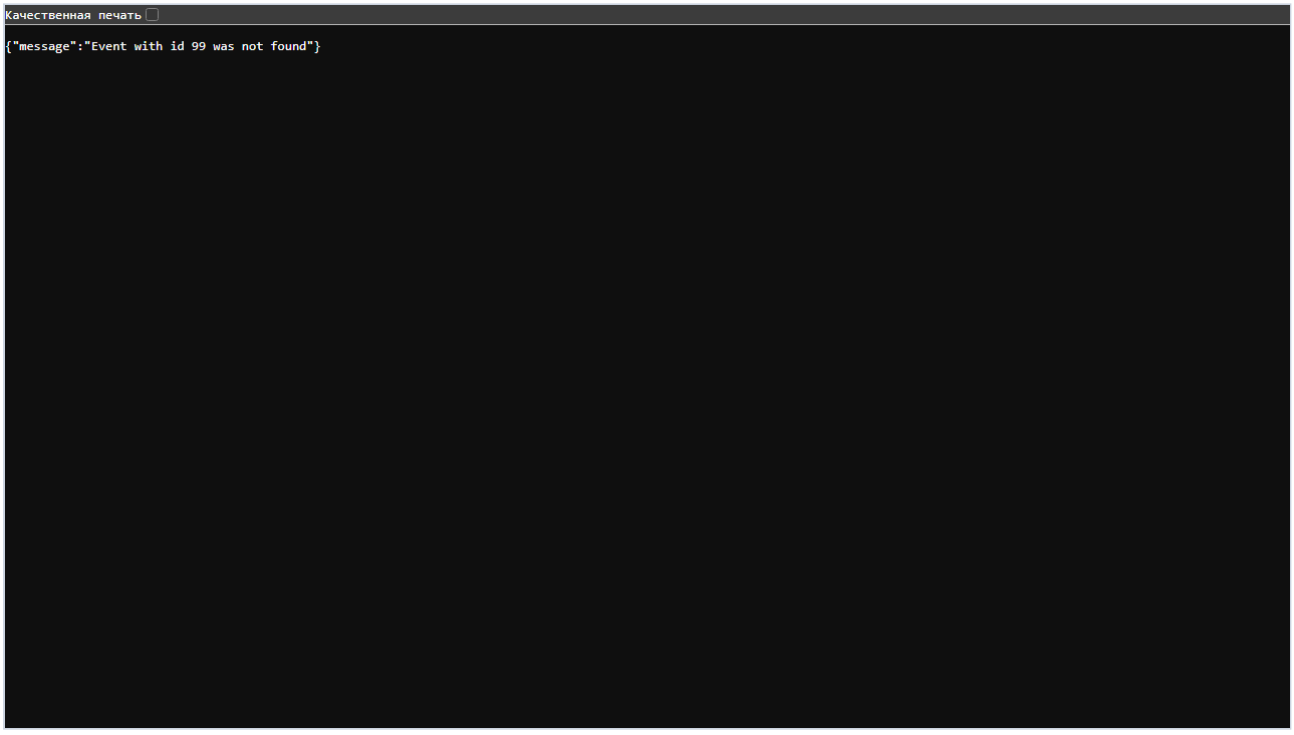


Figure 4. GET /api/events/99 — JSON message for an event ID that does not exist.

Conclusion

This laboratory work introduced the structure of a small Spring Boot REST application. The controller handles HTTP requests while the service provides the welcome message and in-memory event data. Constructor-based dependency injection keeps the controller separate from service creation. The API returned JSON for the required endpoints, and the optional lookup produced a clear 404 response for an unknown event.